

POLYP DETECTION MEDICAL REPORT

Analysis Date: June 04, 2026

VIDEO INFORMATION

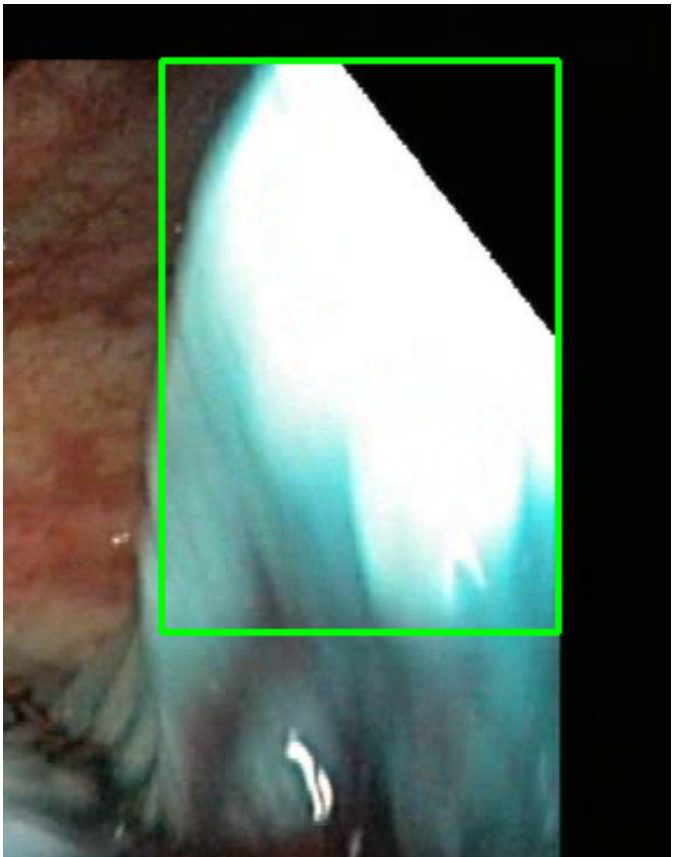
Video ID:	20cdb729-afeb-42c2-b478-b54e24221ecd
Total Frames:	1358
Frame Rate:	25.0 fps (assumed)
Duration:	54.3 seconds

DETECTION SUMMARY

Detection Method	Count	Status
YOLO Detections	157	✓
RT-DETR Detections	1178	✓
MedSAM2 Detections	1036	✓
Consensus (3-Model Agreement)	9	✓
Risk Classification	Count	Percentage
HIGH RISK	0	0.0%
MEDIUM RISK	0	0.0%
LOW RISK	3	33.3%
TOTAL ANALYZED	9	100.0%

DETAILED POLYP ANALYSIS

POLYP #1 — LIFTED_POLYP [LOW_RISK] (Tracked across 29 frames)



Feature	Value	Interpretation
Frame Index	484	Frame 484 of 1358
Detection Method	Detector Consensus (MedSAM2 Unavailable)	MedSAM2 Unavailable (MedSAM2 unavailable for this video)
POLYP TYPE	LIFTED_POLYP	Type confidence: 63.8%
Redness Score	0.020	Color-based vascularization
Relative Radius	23.7%	Polyp size as % of frame diagonal
Texture Score	0.196	Surface morphology
Vessel Visibility	0.009	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification

Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	68.9%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #2 — LIFTED_POLYP [LOW_RISK] (Tracked across 173 frames)



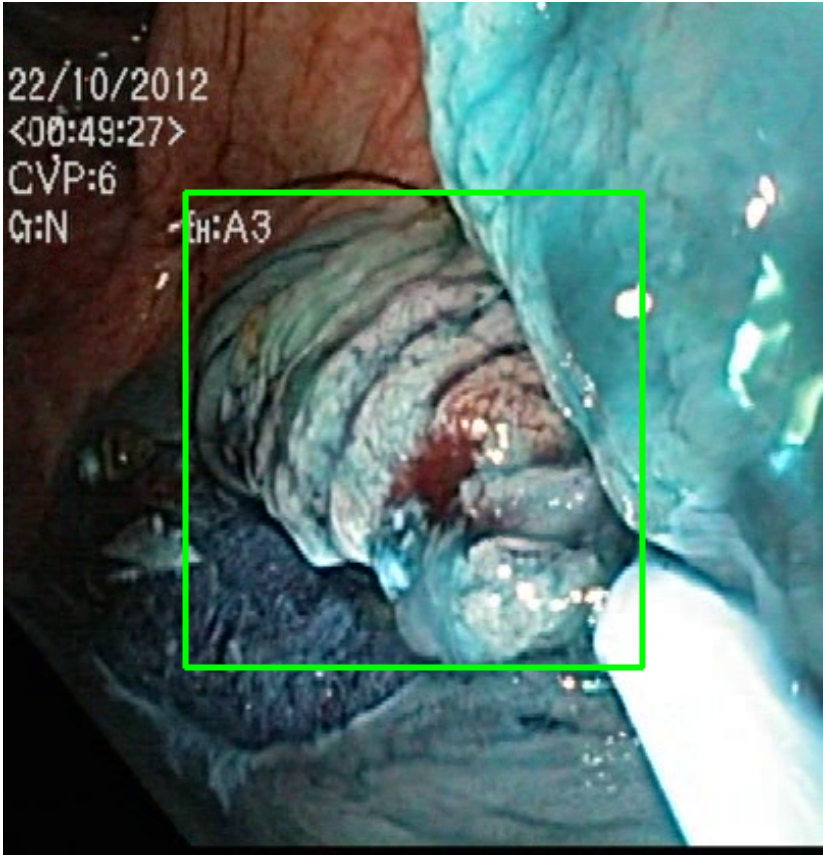
Feature	Value	Interpretation
Frame Index	1010	Frame 1010 of 1358
Detection Method	Detector Consensus (YOLO / RT-DETR)	Model Agreement (MedSAM2 unavailable for this video)
POLYP TYPE	LIFTED_POLYP	Type confidence: 71.3%
Redness Score	0.070	Color-based vascularization
Relative Radius	31.8%	Polyp size as % of frame diagonal
Texture Score	0.173	Surface morphology

Vessel Visibility	0.000	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	67.3%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #3 — LIFTED_POLYP [LOW_RISK] (Tracked across 97 frames)



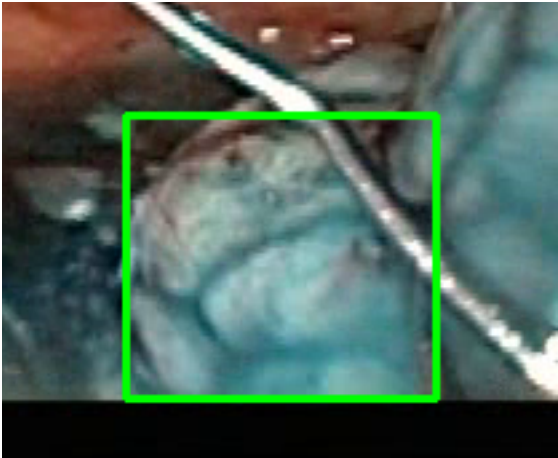
Feature	Value	Interpretation
Frame Index	836	Frame 836 of 1358
Detection Method	Detector Consensus (YOLO / RT-DETR)	Model Agreement (MedSAM2 unavailable for this video)

POLYP TYPE	LIFTED_POLYP	Type confidence: 78.5%
Redness Score	0.000	Color-based vascularization
Relative Radius	12.7%	Polyp size as % of frame diagonal
Texture Score	0.491	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	63.4%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #4 — LIFTED_POLYP [LOW_RISK] (Tracked across 44 frames)

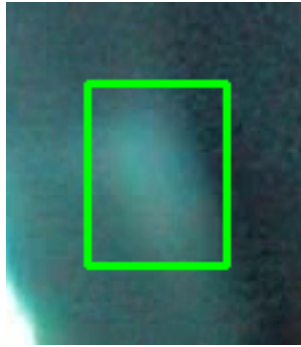


Feature	Value	Interpretation
Frame Index	2	Frame 2 of 1358
Detection Method	Detector Consensus (YOLO / RT-DETR)	Model Agreement (MedSAM2 unavailable for this video)
POLYP TYPE	LIFTED_POLYP	Type confidence: 63.5%
Redness Score	0.012	Color-based vascularization
Relative Radius	12.9%	Polyp size as % of frame diagonal
Texture Score	0.222	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	85.2%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #5 — LIFTED_POLYP [LOW_RISK] (Tracked across 21 frames)



Feature	Value	Interpretation
Frame Index	812	Frame 812 of 1358
Detection Method	Detector Consensus (MedSAM2 Unavailable)	MedSAM2 Unavailable (MedSAM2 unavailable for this video)
POLYP TYPE	LIFTED_POLYP	Type confidence: 64.9%
Redness Score	0.000	Color-based vascularization
Relative Radius	8.0%	Polyp size as % of frame diagonal
Texture Score	0.030	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	NORMAL_MUCOSA	Normal mucosal pattern — no polyp morphology identified
Detection Confidence	73.5%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #6 — LIFTED_POLYP [LOW_RISK] (Tracked across 29 frames)

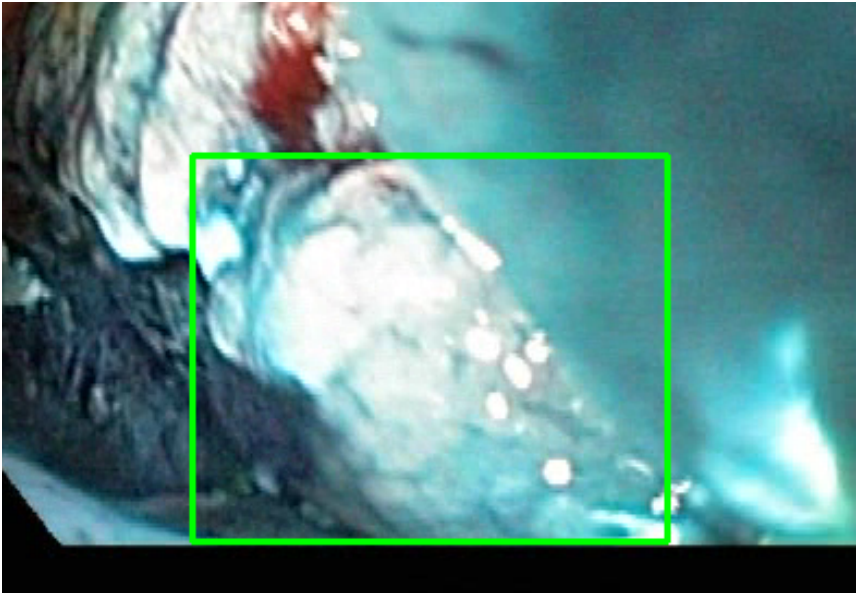


Feature	Value	Interpretation
Frame Index	484	Frame 484 of 1358
Detection Method	Detector Consensus (MedSAM2 Unavailable)	MedSAM2 Unavailable (MedSAM2 unavailable for this video)
POLYP TYPE	LIFTED_POLYP	Type confidence: 63.8%
Redness Score	0.020	Color-based vascularization
Relative Radius	23.7%	Polyp size as % of frame diagonal
Texture Score	0.196	Surface morphology
Vessel Visibility	0.009	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	55.9%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #7 — LIFTED_POLYP [LOW_RISK] (Tracked across 21 frames)

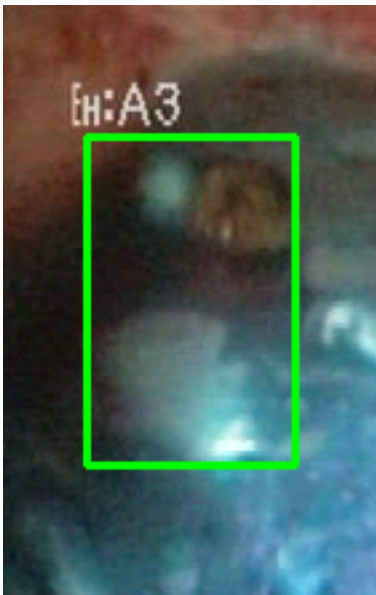


Feature	Value	Interpretation
Frame Index	812	Frame 812 of 1358
Detection Method	Detector Consensus (MedSAM2 Unavailable)	MedSAM2 Unavailable (MedSAM2 unavailable for this video)
POLYP TYPE	LIFTED_POLYP	Type confidence: 64.9%
Redness Score	0.000	Color-based vascularization
Relative Radius	8.0%	Polyp size as % of frame diagonal
Texture Score	0.030	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	NORMAL_MUCOSA	Normal mucosal pattern — no polyp morphology identified
Detection Confidence	75.4%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

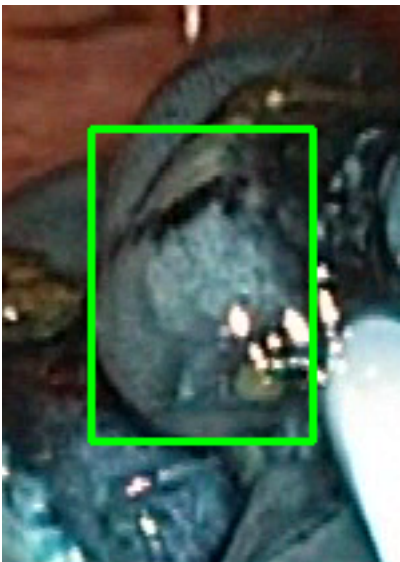
POLYP #8 — LIFTED_POLYP [LOW_RISK] (Tracked across 17 frames)



Feature	Value	Interpretation
Frame Index	468	Frame 468 of 1358
Detection Method	Detector Consensus (MedSAM2 Unavailable)	Model Agreement (MedSAM2 unavailable for this video)
POLYP TYPE	LIFTED_POLYP	Type confidence: 63.9%
Redness Score	0.023	Color-based vascularization
Relative Radius	37.2%	Polyp size as % of frame diagonal
Texture Score	0.153	Surface morphology
Vessel Visibility	0.037	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	53.6%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

POLYP #9 — LIFTED_POLYP [LOW_RISK] (Tracked across 14 frames)



Feature	Value	Interpretation
Frame Index	951	Frame 951 of 1358
Detection Method	Detector Consensus (MedSAM2 Unavailable)	MedSAM2 Unavailable (MedSAM2 unavailable for this video)
POLYP TYPE	LIFTED_POLYP	Type confidence: 77.4%
Redness Score	0.014	Color-based vascularization
Relative Radius	12.7%	Polyp size as % of frame diagonal
Texture Score	0.507	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	58.1%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

CLINICAL IMPRESSION

Summary:

A total of 9 polyps were detected and analyzed using multi-model consensus and AI-based risk stratification.

Risk Distribution:

- HIGH RISK polyps: 0 (0.0%)
- MEDIUM RISK polyps: 0 (0.0%)
- LOW RISK polyps: 3 (33.3%)

Recommendation:

All detected polyps are classified as LOW RISK. Standard surveillance protocol is recommended.

Technical Details:

- Detection: Multi-model consensus (YOLO, RT-DETR, MedSAM2)
- Features: 444-dimensional vectors (384 SSL + 60 biomarkers)
- Classification: Cluster-specific Decision Tree experts
- Confidence: Based on deep learning + medical heuristics

Disclaimer: This report is generated by an AI-assisted research system for academic and research purposes only. All clinical recommendations are derived from published ESGE 2017, NICE CG118/CG131, and BSG 2019 colonoscopy guidelines. This output does not constitute medical advice and must not be used as the sole basis for clinical decisions. All findings must be reviewed and confirmed by a qualified gastroenterologist or endoscopist.